

```
In [3]: import boto3
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
import io, boto3
```

```
In [4]: s3 = boto3.client('s3')
bucket = 'cust-seg-ygp' #bucket name
key = 'online_retail_II.csv' #file inside the bucket

obj = s3.get_object(Bucket=bucket, Key=key)
df = pd.read_csv(obj['Body'], low_memory=False)
print("Loaded rows:", len(df))
df.head()
```

Loaded rows: 1067371

```
Out[4]:
```

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-12-01 07:45:00	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-12-01 07:45:00	2.10	13085.0	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-12-01 07:45:00	1.25	13085.0	United Kingdom

```
In [5]: print("Shape (rows, columns):", df.shape)
```

Shape (rows, columns): (1067371, 8)

```
In [6]: print("Total rows:", df.shape[0])
```

Total rows: 1067371

```
In [7]: print("Column names:", df.columns.tolist())
```

Column names: ['Invoice', 'StockCode', 'Description', 'Quantity', 'InvoiceDate', 'Price', 'Customer ID', 'Country']

```
In [8]: print(df.dtypes)
```

```

Invoice      object
StockCode    object
Description   object
Quantity     int64
InvoiceDate  object
Price        float64
Customer ID  float64
Country      object
dtype: object

```

```
In [9]: print("Number of duplicate rows:", df.duplicated().sum())
```

Number of duplicate rows: 34335

```
In [10]: # Show duplicate rows
dup_rows = df[df.duplicated()]
print(dup_rows.head())
```

	Invoice	StockCode	Description	Quantity	\
371	489517	21912	VINTAGE SNAKES & LADDERS	1	
383	489517	22130	PARTY CONE CHRISTMAS DECORATION	6	
384	489517	22319	HAIRCLIPS FORTIES FABRIC ASSORTED	12	
385	489517	21913	VINTAGE SEASIDE JIGSAW PUZZLES	1	
386	489517	21821	GLITTER STAR GARLAND WITH BELLS	1	

	InvoiceDate	Price	Customer ID	Country
371	2009-12-01 11:34:00	3.75	16329.0	United Kingdom
383	2009-12-01 11:34:00	0.85	16329.0	United Kingdom
384	2009-12-01 11:34:00	0.65	16329.0	United Kingdom
385	2009-12-01 11:34:00	3.75	16329.0	United Kingdom
386	2009-12-01 11:34:00	3.75	16329.0	United Kingdom

```
In [11]: df = df.drop_duplicates()
```

```
In [12]: print("After removing duplicates:", df.shape)
```

After removing duplicates: (1033036, 8)

```
In [13]: print(df.isnull().sum())
```

```

Invoice      0
StockCode    0
Description   4275
Quantity     0
InvoiceDate  0
Price        0
Customer ID  235151
Country      0
dtype: int64

```

```
In [14]: # View missing Customer ID rows BEFORE deleting
print(df['Customer ID'].isna().sum())
print(df[df['Customer ID'].isna()].head(5))
```

235151

	Invoice	StockCode	Description	Quantity	\
263	489464	21733	85123a mixed	-96	
283	489463	71477	short	-240	
284	489467	85123A	21733 mixed	-192	
470	489521	21646	NaN	-50	
577	489525	85226C	BLUE PULL BACK RACING CAR	1	

	InvoiceDate	Price	Customer ID	Country
263	2009-12-01 10:52:00	0.00	NaN	United Kingdom
283	2009-12-01 10:52:00	0.00	NaN	United Kingdom
284	2009-12-01 10:53:00	0.00	NaN	United Kingdom
470	2009-12-01 11:44:00	0.00	NaN	United Kingdom
577	2009-12-01 11:49:00	0.55	NaN	United Kingdom

```
In [15]: # Drop rows with missing Customer ID
df = df.dropna(subset=["Customer ID"])
print("After dropping missing Customer ID rows:", df.shape)
```

After dropping missing Customer ID rows: (797885, 8)

```
In [16]: # Convert Customer ID from float to int
df["Customer ID"] = df["Customer ID"].astype(int)
```

```
In [17]: # View negative quantities BEFORE dropping
df[df['Quantity'] < 0].head(4)
```

```
Out[17]:
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	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
178	C489449	22087	PAPER BUNTING WHITE LACE	-12	2009-12-01 10:33:00	2.95	16321	Australia
179	C489449	85206A	CREAM FELT EASTER EGG BASKET	-6	2009-12-01 10:33:00	1.65	16321	Australia
180	C489449	21895	POTTING SHED SOW 'N' GROW SET	-4	2009-12-01 10:33:00	4.25	16321	Australia
181	C489449	21896	POTTING SHED TWINE	-6	2009-12-01 10:33:00	2.10	16321	Australia

```
In [18]: # Remove negative quantities (returns)
df = df[df['Quantity'] > 0]
```

```
In [19]: # View negative quantities AFTER dropping
df[df['Quantity'] < 0].head(4)
```

```
Out[19]:
```

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
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```
In [20]: # Remove zero/negative prices
print((df['Price'] <= 0).sum())
print(df[df['Price'] <= 0].head(5))
```

70

	Invoice	StockCode	Description	Quantity	\
4674	489825	22076	6 RIBBONS EMPIRE	12	
6781	489998	48185	DOOR MAT FAIRY CAKE	2	
16107	490727	M	Manual	1	
18738	490961	22065	CHRISTMAS PUDDING TRINKET POT	1	
18739	490961	22142	CHRISTMAS CRAFT WHITE FAIRY	12	

	InvoiceDate	Price	Customer ID	Country
4674	2009-12-02 13:34:00	0.0	16126	United Kingdom
6781	2009-12-03 11:19:00	0.0	15658	United Kingdom
16107	2009-12-07 16:38:00	0.0	17231	United Kingdom
18738	2009-12-08 15:25:00	0.0	14108	United Kingdom
18739	2009-12-08 15:25:00	0.0	14108	United Kingdom

```
In [21]: df = df[df["Price"] > 0]
```

```
In [22]: # AFTER Removing zero/negative prices
print((df['Price'] <= 0).sum())
print(df[df['Price'] <= 0].head(10))
```

```
0
Empty DataFrame
Columns: [Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer
ID, Country]
Index: []
```

```
In [23]: # InvoiceDate is a datetime
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
```

```
In [24]: import io

# your bucket name
bucket_name = "cust-seg-ygp"

# your output filename in S3
file_name = "cleaned_online_retail_II.csv"

# convert dataframe to in-memory CSV
csv_buffer = io.StringIO()
df.to_csv(csv_buffer, index=False)

# upload to S3
s3 = boto3.resource('s3')
s3.Object(bucket_name, file_name).put(Body=csv_buffer.getvalue())

print(f"Saved cleaned file to s3://{bucket_name}/{file_name}")

Saved cleaned file to s3://cust-seg-ygp/cleaned_online_retail_II.csv
```

```
In [25]: df.head(5)
```

Out[25]:

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
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